

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5093B

5 Credits: 4

Program Elective

Source-grounded

Speciality Polymers

□ To impart knowledge on - high performance and advanced technological applications

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5093B is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5093B.pdf from the uploaded official SITTTTR syllabus archive.

Course code and title: preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Polymer Technology
Course code	5093B
Course title	Speciality Polymers
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4,T:0,P:0) Periods per semester: 60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES**Objectives, outcomes and mapping****Course objectives**

- To impart knowledge on - high performance and advanced technological applications
- of polymers.

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding of high temperature and fire-resistant polymers. Describe the structure, properties and	See official syllabus
CO2	application of polymers used in electrical and 15 Understanding electronics fields. Explain the properties and application of	See official syllabus
CO3	polymers used in construction, biomedical, 14 Understanding aerospace and telecommunication Describe the properties and application of ionomers, hydrogel, nano materials and nano 14 Understanding	See official syllabus
CO4	composites.	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	and fire-resistant polymers	4	As specified
Module 2	electrical and electronics fields.	4	As specified
Module 3	telecommunication	4	As specified
Module 4	and nano materials.	4	As specified

MODULE 1**and fire-resistant polymers**

This module is presented from the official syllabus scope for Speciality Polymers. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: and fire-resistant polymers
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 2 Remembering stability and fire resistance of polymers. Explain methods for improving the thermal

2 Remembering stability and fire resistance of polymers. Explain methods for improving the thermal is an official topic in Module 1 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Remembering stability and fire resistance of polymers. Explain methods for improving the thermal using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 remembering stability and fire resistance of polymers. explain methods for improving the thermal should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 2 Remembering stability and fire resistance of polymers. Explain methods for improving the thermal means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Remembering stability and fire resistance of polymers. Explain methods for improving the thermal
 definition/meaning .
 , steps, application . Technical terms
 English- .

▼ 2 Remembering stability and flame resistance of polymers. Describe the properties and high temperature

2 Remembering stability and flame resistance of polymers. Describe the properties and high temperature is an official topic in Module 1 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Remembering stability and flame resistance of polymers. Describe the properties and high temperature using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 remembering stability and flame resistance of polymers. describe the properties and high temperature should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Remembering stability and flame resistance of polymers. Describe the properties and high temperature means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Remembering stability and flame resistance of polymers. Describe the properties and high temperature
 definition/meaning .
 , steps, application . Technical terms
 English- .

▼ applications of polymers-PTFE, Aromatic 6 Understanding Polymers - Poly phenylene Sulphide, PPO. Explain the properties and applications of

applications of polymers-PTFE, Aromatic 6 Understanding Polymers -Poly phenylene Sulphide, PPO. Explain the properties and applications of is an official topic in Module 1 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify applications of polymers-PTFE, Aromatic 6 Understanding Polymers -Poly phenylene Sulphide, PPO. Explain the properties and applications of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, applications of polymers-ptfe, aromatic 6 understanding polymers -poly phenylene sulphide, ppo. explain the properties and applications of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What applications of polymers-PTFE, Aromatic 6 Understanding Polymers -Poly phenylene Sulphide, PPO. Explain the properties and applications of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പൊതുവായ പൊളിമറുകളുടെ പ്രയോഗങ്ങൾ-PTFE, Aromatic 6 Understanding Polymers -Poly phenylene Sulphide, PPO. Explain the properties and applications of പൊളിമറുകൾ. പൊളിമറുകളുടെ definition/meaning പൊളിമറുകളുടെ പ്രയോഗങ്ങൾ. പൊളിമറുകളുടെ പ്രയോഗങ്ങൾ, steps, application പൊളിമറുകളുടെ പ്രയോഗങ്ങൾ. Technical terms English-പൊളിമറുകൾ പ്രയോഗങ്ങൾ.

▼ PEEK, Aramid fibres, Heterocyclic 5 Understanding polymers-PBI High temperature and fire-resistant polymers improving low performance polymers for high temperature use - polymers for low fire hazards - polymers for high temperature resistance. Factors affecting the thermal stability and fire resistance of polymers - LOI- Definition Properties and applications of PTFE, Aromatic Polymers like Poly phenylene Sulphide, PPO, PEEK, Aramid, Heterocyclic polymers - polybenzimidazole (PBI) Describe the structure, properties and application of polymers used in

PEEK, Aramid fibres, Heterocyclic 5 Understanding polymers-PBI High temperature and fire-resistant polymers improving low performance polymers for high temperature use - polymers for low fire hazards - polymers for high temperature resistance. Factors affecting the thermal stability and fire resistance of polymers - LOI- Definition Properties and applications of PTFE, Aromatic Polymers like Poly phenylene Sulphide, PPO, PEEK, Aramid, Heterocyclic polymers - polybenzimidazole (PBI) Describe the structure, properties and application of polymers used in is an official topic in Module 1 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify PEEK, Aramid fibres, Heterocyclic 5 Understanding polymers-PBI High temperature and fire-resistant polymers improving low performance polymers for high temperature use - polymers for low fire hazards - polymers for high temperature resistance. Factors affecting the thermal stability and fire resistance of polymers - LOI- Definition Properties

and applications of PTFE, Aromatic Polymers like Poly phenylene Sulphide, PPO, PEEK, Aramid, Heterocyclic polymers - polybenzimidazole (PBI)

Describe the structure, properties and application of polymers used in using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, peek, aramid fibres, heterocyclic 5 understanding polymers-pbi high temperature and fire-resistant polymers improving low performance polymers for high temperature use - polymers for low fire hazards - polymers for high temperature resistance. factors affecting the thermal stability and fire resistance of polymers – loi- definition properties and applications of ptfe, aromatic polymers like poly phenylene sulphide, ppo, peek, aramid, heterocyclic polymers - polybenzimidazole (pbi) describe the structure, properties and application of polymers used in should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What PEEK, Aramid fibres, Heterocyclic 5 Understanding polymers-PBI High temperature and fire-resistant polymers improving low performance polymers for high temperature use - polymers for low fire hazards - polymers for high temperature resistance. Factors affecting the thermal stability and fire resistance of polymers – LOI- Definition Properties and applications of PTFE, Aromatic Polymers like Poly phenylene Sulphide, PPO, PEEK, Aramid, Heterocyclic polymers - polybenzimidazole (PBI) Describe the structure, properties and application of polymers used in means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- PEEK, Aramid fibres, Heterocyclic 5 Understanding polymers-PBI High temperature and fire-resistant polymers improving low performance polymers for high temperature use - polymers for low fire hazards - polymers for high temperature resistance. Factors affecting the thermal stability and fire resistance of polymers - LOI- Definition Properties and applications of PTFE, Aromatic Polymers like Poly phenylene Sulphide, PPO, PEEK, Aramid, Heterocyclic polymers - polybenzimidazole (PBI) Describe the structure, properties and application of polymers used in definition/meaning. steps, application Technical terms English-

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2**electrical and electronics fields.**

This module is presented from the official syllabus scope for Speciality Polymers. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: electrical and electronics fields.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 2 Remembering conducting mechanism. Describe the structure, properties and application of conducting polymers like

2 Remembering conducting mechanism. Describe the structure, properties and application of conducting polymers like is an official topic in Module 2 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Remembering conducting mechanism. Describe the structure, properties and application of conducting polymers like using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 remembering conducting mechanism. describe the structure, properties and application of conducting polymers like should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Remembering conducting mechanism. Describe the structure, properties and application of conducting polymers like means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 2 Remembering conducting mechanism. Describe the structure, properties and application of conducting polymers like definition/meaning. , steps, application . Technical terms English- .

▼ 5 Understanding Polyacetylene, poly aniline, poly paraphenylene

Describe the properties and application of

5 Understanding Polyacetylene, poly aniline, poly paraphenylene Describe the properties and application of is an official topic in Module 2 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding Polyacetylene, poly aniline, poly paraphenylene Describe the properties and application of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding polyacetylene, poly aniline, poly paraphenylene describe the properties and application of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding Polyacetylene, poly aniline, poly paraphenylene Describe the properties and application of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- 5 Understanding Polyacetylene, poly aniline, poly paraphenylene Describe the properties and application of definition/meaning. , steps, application . Technical terms English- .

▼ photo conducting and piezo electric polymers - 4 Understanding Poly vinyl carbazole , PVDF Explain the properties and application of Liquid crystalline polymers and Photo

photo conducting and piezo electric polymers - 4 Understanding Poly vinyl carbazole , PVDF Explain the properties and application of Liquid crystalline polymers and Photo is an official topic in Module 2 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify photo conducting and piezo electric polymers - 4 Understanding Poly vinyl carbazole , PVDF Explain the properties and application of Liquid crystalline polymers and Photo using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, photo conducting and piezo electric polymers - 4 understanding poly vinyl carbazole , pvdf explain the properties and application of liquid crystalline polymers and photo should be discussed with attention to

safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What photo conducting and piezo electric polymers - 4 Understanding Poly vinyl carbazole , PVDF Explain the properties and application of Liquid crystalline polymers and Photo means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-പ്രകാശ സഞ്ചാലകതയും പൈയോ ഇലക്ട്രിക് പോളിമറുകളും - 4 Understanding Poly vinyl carbazole , PVDF Explain the properties and application of Liquid crystalline polymers and Photo പ്രകാശ സഞ്ചാലകതയും പൈയോ ഇലക്ട്രിക് പോളിമറുകളും definition/meaning പ്രകാശം പ്രകാശം പ്രകാശം, steps, application പ്രകാശം പ്രകാശം പ്രകാശം. Technical terms English-പ്രകാശം പ്രകാശം പ്രകാശം.

▼ 4 Understanding resists polymers for semiconductor application. Polymers with electrical and electronic properties Conducting polymers -definition conducting mechanisms Properties and application of Polyacetylene, Polyparaphenylene, and poly aniline. Photo conducting polymers - structure, properties application of Poly vinyl carbazole Piezo electric polymers properties - structure, properties and application PVDF Photo resist polymers for semi conductor fabrication - Liquid crystalline polymers. Explain the polymers used in construction, biomedical, aerospace and

4 Understanding resists polymers for semiconductor application. Polymers with electrical and electronic properties Conducting polymers -definition conducting mechanisms Properties and application of Polyacetylene, Polyparaphenylene, and poly aniline. Photo conducting polymers - structure, properties application of Poly vinyl carbazole Piezo electric polymers properties - structure, properties and application PVDF Photo resist polymers for semi conductor fabrication - Liquid crystalline polymers. Explain the polymers used in construction, biomedical, aerospace and is an official topic in Module 2 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding resists polymers for semiconductor application. Polymers with electrical and electronic properties Conducting polymers -definition conducting mechanisms Properties and application of Polyacetylene, Polyparaphenylene, and poly aniline. Photo conducting

polymers - structure, properties application of Poly vinyl carbazole Piezo electric polymers properties - structure, properties and application PVDF Photo resist polymers for semi conductor fabrication - Liquid crystalline polymers. Explain the polymers used in construction, biomedical, aerospace and using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding resists polymers for semiconductor application. polymers with electrical and electronic properties conducting polymers -definition conducting mechanisms properties and application of polyacetylene, polyparaphenylene, and poly aniline. photo conducting polymers - structure, properties application of poly vinyl carbazole piezo electric polymers properties - structure, properties and application pvdf photo resist polymers for semi conductor fabrication - liquid crystalline polymers. explain the polymers used in construction, biomedical, aerospace and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding resists polymers for semiconductor application. Polymers with electrical and electronic properties Conducting polymers - definition conducting mechanisms Properties and application of Polyacetylene, Polyparaphenylene, and poly aniline. Photo conducting polymers - structure, properties application of Poly vinyl carbazole Piezo electric polymers properties - structure, properties and application PVDF Photo resist polymers for semi conductor fabrication - Liquid crystalline polymers. Explain the polymers used in construction, biomedical, aerospace and means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-4 Understanding resists polymers for semiconductor application. Polymers with electrical and electronic properties Conducting polymers -definition conducting mechanisms Properties and application of Polyacetylene, Polyparaphenylene, and poly aniline. Photo conducting polymers - structure, properties application of Poly vinyl carbazole Piezo electric polymers properties - structure, properties and application PVDF Photo resist polymers for semi conductor fabrication - Liquid crystalline polymers. Explain the polymers used in construction, biomedical, aerospace and definition/meaning. steps, application. Technical terms English-

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

telecommunication

This module is presented from the official syllabus scope for Speciality Polymers. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: telecommunication
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ Explain the polymer concrete, types. 3 Understanding Describe the polymers and its properties

Explain the polymer concrete, types. 3 Understanding Describe the polymers and its properties is an official topic in Module 3 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the polymer concrete, types. 3 Understanding Describe the polymers and its properties using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the polymer concrete, types. 3 understanding describe the polymers and its properties should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Explain the polymer concrete, types. 3 Understanding Describe the polymers and its properties means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain the polymer concrete, types. 3 Understanding Describe the polymers and its properties definition/meaning . steps, application . Technical terms English- .

▼ 5 Understanding used for biomedical application. Explain the properties of polymers used as

5 Understanding used for biomedical application. Explain the properties of polymers used as is an official topic in Module 3 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding used for biomedical application. Explain the properties of polymers used as using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding used for biomedical application. explain the properties of polymers used as should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
--------	-----------------

Aspect	What to prepare
Definition/identity	What 5 Understanding used for biomedical application. Explain the properties of polymers used as means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-5 Understanding used for biomedical application. Explain the properties of polymers used as definition/meaning. steps, application. Technical terms English-.

▼ 3 Understanding binders for rocket propellants. Explain polymers used for Optical fibre for

3 Understanding binders for rocket propellants. Explain polymers used for Optical fibre for is an official topic in Module 3 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding binders for rocket propellants. Explain polymers used for Optical fibre for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding binders for rocket propellants. explain polymers used for optical fibre for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 3 Understanding binders for rocket propellants. Explain polymers used for Optical fibre for means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 3 Understanding binders for rocket propellants. Explain polymers used for Optical fibre for definition/meaning, steps, application Technical terms English-

▼ 3 Understanding telecommunication cables (POFC). Polymer concrete-preparation properties, application, - PCC, PIC, PMC. Polymers for biomedical applications - PHEMA, SILICONE - properties, application Polymeric binders for rocket propellants - HTPB - properties Polymers used in Optical fibre Cable(POFC) - manufacturing, properties and application Explain the ionomers, hydrogel, Biodegradable polymers, nano polymers

3 Understanding telecommunication cables (POFC). Polymer concrete-preparation properties, application, - PCC, PIC, PMC. Polymers for biomedical applications - PHEMA, SILICONE - properties, application Polymeric binders for rocket propellants - HTPB - properties Polymers used in Optical fibre Cable(POFC) - manufacturing, properties and application Explain the ionomers, hydrogel, Biodegradable polymers, nano polymers is an official topic in Module 3 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding telecommunication cables (POFC). Polymer concrete- preparation properties, application, - PCC, PIC, PMC. Polymers for biomedical applications - PHEMA, SILICONE - properties, application Polymeric binders for rocket propellants - HTPB - properties Polymers used in Optical fibre Cable(POFC) - manufacturing, properties and application Explain the ionomers, hydrogel, Biodegradable polymers, nano polymers using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding telecommunication cables (pofc). polymer concrete- preparation properties, application, - pcc, pic, pmc. polymers for biomedical applications - phema, silicone - properties, application polymeric binders for rocket propellants - httpb - properties polymers used in optical fibre cable(pofc) – manufacturing, properties and application explain the ionomers, hydrogel, biodegradable polymers, nano polymers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding telecommunication cables (POFC). Polymer concrete- preparation properties, application, - PCC, PIC, PMC. Polymers for biomedical applications - PHEMA, SILICONE - properties, application Polymeric binders for rocket propellants - HTPB - properties Polymers used in Optical fibre Cable(POFC) – manufacturing, properties and application Explain the ionomers, hydrogel, Biodegradable polymers, nano polymers means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 3 Understanding telecommunication cables (POFC). Polymer concrete- preparation properties, application, - PCC, PIC, PMC. Polymers for biomedical applications - PHEMA, SILICONE - properties, application Polymeric binders for rocket propellants - HTPB - properties Polymers used in Optical fibre Cable(POFC) – manufacturing, properties and application Explain the ionomers, hydrogel, Biodegradable polymers, nano polymers definition/meaning . steps, application . Technical terms English- .

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4**and nano materials.**

This module is presented from the official syllabus scope for Speciality Polymers. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: and nano materials.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 2 Remembering electrolytes and hydrogel. Describe the Synthesis, physical

2 Remembering electrolytes and hydrogel. Describe the Synthesis, physical is an official topic in Module 4 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Remembering electrolytes and hydrogel. Describe the Synthesis, physical using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 remembering electrolytes and hydrogel. describe the synthesis, physical should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 2 Remembering electrolytes and hydrogel. Describe the Synthesis, physical means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 2 Remembering electrolytes and hydrogel. Describe the Synthesis, physical definition/meaning. , steps, application . Technical terms English- .

▼ properties and applications of ionomers 4 Understanding based on PS, PTFE Explain the properties and applications of

properties and applications of ionomers 4 Understanding based on PS, PTFE Explain the properties and applications of is an official topic in Module 4 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify properties and applications of ionomers 4 Understanding based on PS, PTFE Explain the properties and applications of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, properties and applications of ionomers 4 understanding based on ps, ptfe explain the properties and applications of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What properties and applications of ionomers 4 Understanding based on PS, PTFE Explain the properties and applications of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഇയ്യ പ്രപർട്ടീസ് ആന്റ് അപ്ലിക്കേഷൻസ് ഓഫ് ഓണമർ 4 Understanding based on PS, PTFE Explain the properties and applications of ഇയ്യ പ്രപർട്ടീസ് ആന്റ് അപ്ലിക്കേഷൻസ് definition/meaning ഇയ്യ പ്രപർട്ടീസ് ആന്റ് അപ്ലിക്കേഷൻസ്. ഇയ്യ പ്രപർട്ടീസ് ആന്റ് അപ്ലിക്കേഷൻസ്, steps, application ഇയ്യ പ്രപർട്ടീസ് ആന്റ് അപ്ലിക്കേഷൻസ്. Technical terms English-ഇയ്യ പ്രപർട്ടീസ് ആന്റ് അപ്ലിക്കേഷൻസ്.

▼ 4 Understanding biodegradable polymers, PLA, Chitosan. Generalize the idea of nano polymers,

4 Understanding biodegradable polymers, PLA, Chitosan. Generalize the idea of nano polymers, is an official topic in Module 4 of Speciality Polymers. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding biodegradable polymers, PLA, Chitosan. Generalize the idea of nano polymers, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding biodegradable polymers, pla, chitosan. generalize the idea of nano polymers, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 4 Understanding biodegradable polymers, PLA, Chitosan. Generalize the idea of nano polymers, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- 4 Understanding biodegradable polymers, PLA, Chitosan. Generalize the idea of nano polymers, definition/meaning. Steps, application. Technical terms English- Malayalam.

▼ preparation, properties and application of 4 Understanding nano composites Properties of ionic polymers Ion - exchange, Hydrophilicity. Poly electrolytes- definition Synthesis, physical properties and applications of ionomers based on PS, PTFE, Biodegradable polymers - Properties and applications of PLA, chitosan. Polymer Nano composite - Definition, properties and application of nano composites Text / Reference: T/R Book Title/Author H.F.Mark, (Ed), Encyclopedia of polymer Science & Engineering, John Wiley & T1 Sons, New York, 1989. R1 Matrin.T.Goosey, Plastics for Electronics, Elsevier, Applied Science, 1985. Manas Chanda, Salil.K.Roy, Plastics Technology Hand book, 2nd edition, Marcel R2 Dekker, New York, 1993 R3 R.W. Dyson, Specialty Polymers, Chapman & Hall, 2nd edition, 1998. Joseph H Koo, Fundamentals properties and applications of polymer nano R4 composites, Cambridge University Press, 2016. Online Resources: Sl.No Website Link 1 <https://www.researchgate.net/> 2 <https://onlinelibrary.wiley.com> 3 <https://www.ncbi.nlm.nih.gov>

preparation, properties and application of 4 Understanding nano composites Properties of ionic polymers Ion - exchange, Hydrophilicity. Poly electrolytes- definition Synthesis, physical properties and applications of ionomers based on PS, PTFE, Biodegradable polymers - Properties and applications of PLA, chitosan. Polymer Nano composite - Definition, properties and application of nano composites Text / Reference: T/R Book Title/Author H.F.Mark, (Ed), Encyclopedia of polymer Science & Engineering, John Wiley & T1 Sons, New York, 1989. R1 Matrin.T.Goosey, Plastics for Electronics, Elsevier, Applied Science, 1985. Manas Chanda, Salil.K.Roy, Plastics Technology Hand book, 2nd edition, Marcel R2 Dekker, New York, 1993 R3 R.W. Dyson, Specialty Polymers, Chapman & Hall, 2nd edition, 1998. Joseph H Koo, Fundamentals properties and applications of polymer nano R4 composites, Cambridge University Press, 2016. Online Resources: Sl.No Website Link 1 <https://www.researchgate.net/> 2 <https://onlinelibrary.wiley.com> 3 <https://www.ncbi.nlm.nih.gov> is an official topic in Module 4 of Speciality Polymers. Study the terminology first, then

connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify preparation, properties and application of 4
Understanding nano composites Properties of ionic polymers Ion - exchange, Hydrophilicity. Poly electrolytes- definition Synthesis, physical properties and applications of ionomers based on PS, PTFE, Biodegradable polymers - Properties and applications of PLA, chitosan. Polymer Nano composite - Definition, properties and application of nano composites Text / Reference: T/R Book Title/Author H.F.Mark, (Ed), Encyclopedia of polymer Science & Engineering, John Wiley & T1 Sons, New York, 1989. R1 Matrin.T.Goosey, Plastics for Electronics, Elsevier, Applied Science, 1985. Manas Chanda, Salil.K.Roy, Plastics Technology Hand book, 2nd edition, Marcel R2 Dekker, New York, 1993 R3 R.W. Dyson, Specialty Polymers, Chapman & Hall, 2nd edition, 1998. Joseph H Koo, Fundamentals properties and applications of polymer nano R4 composites, Cambridge University Press, 2016. Online Resources: Sl.No Website Link 1 <https://www.researchgate.net/> 2 <https://onlinelibrary.wiley.com> 3 <https://www.ncbi.nlm.nih.gov> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, preparation, properties and application of 4 understanding nano composites properties of ionic polymers ion - exchange, hydrophilicity. poly electrolytes- definition synthesis, physical properties and applications of ionomers based on ps, ptfe, biodegradable polymers - properties and applications of pla, chitosan. polymer nano composite - definition, properties and application of nano composites text / reference: t/r book title/author h.f.mark, (ed), encyclopedia of polymer science & engineering, john wiley & t1 sons, new york, 1989. r1 matrin.t.goosey, plastics for electronics, elsevier, applied science, 1985. manas chanda, salil.k.roy, plastics technology hand book, 2nd edition, marcel r2 dekker, new york, 1993 r3 r.w. dyson, specialty polymers, chapman & hall, 2nd edition, 1998. joseph h koo, fundamentals properties and applications of polymer nano r4 composites, cambridge university press, 2016. online resources: sl.no website link 1 <https://www.researchgate.net/> 2 <https://onlinelibrary.wiley.com> 3 <https://www.ncbi.nlm.nih.gov> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What preparation, properties and application of 4 Understanding nano composites Properties of ionic polymers Ion - exchange, Hydrophilicity. Poly electrolytes- definition Synthesis, physical properties and applications of ionomers based on PS, PTFE, Biodegradable polymers - Properties and applications of PLA, chitosan. Polymer Nano composite - Definition, properties and application of nano composites Text / Reference: T/R Book Title/Author H.F.Mark, (Ed), Encyclopedia of polymer Science & Engineering, John Wiley & T1 Sons, New York, 1989. R1 Matrin.T.Goosey, Plastics for Electronics, Elsevier, Applied Science, 1985. Manas Chanda, Salil.K.Roy, Plastics Technology Hand book, 2nd edition, Marcel R2 Dekker, New York, 1993 R3 R.W. Dyson, Specialty Polymers, Chapman & Hall, 2nd edition, 1998. Joseph H Koo, Fundamentals properties and applications of polymer nano R4 composites, Cambridge University Press, 2016. Online Resources: Sl.No Website Link 1 https://www.researchgate.net/ 2 https://onlinelibrary.wiley.com 3 https://www.ncbi.nlm.nih.gov means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-നാണുവസ്തുതയുടെ തയ്യാറെടുപ്പ്, ഗുണങ്ങൾ, പ്രയോഗം of 4 Understanding nano composites Properties of ionic polymers Ion - exchange, Hydrophilicity. Poly electrolytes- definition Synthesis, physical properties and applications of ionomers based on PS, PTFE, Biodegradable polymers - Properties and applications of PLA, chitosan. Polymer Nano composite - Definition, properties and application of nano composites Text / Reference: T/R Book Title/Author H.F.Mark, (Ed), Encyclopedia of polymer Science & Engineering, John Wiley & T1 Sons, New York, 1989. R1 Matrin.T.Goosey, Plastics for Electronics, Elsevier, Applied Science, 1985. Manas Chanda, Salil.K.Roy, Plastics Technology Hand book, 2nd edition, Marcel R2 Dekker, New York, 1993 R3 R.W. Dyson, Specialty Polymers, Chapman & Hall, 2nd edition, 1998. Joseph H Koo, Fundamentals properties and applications of polymer nano R4 composites, Cambridge University Press, 2016. Online Resources: Sl.No Website Link 1 <https://www.researchgate.net/> 2 <https://onlinelibrary.wiley.com> 3 <https://www.ncbi.nlm.nih.gov> നാണുവസ്തുതയുടെ നിർവ്വചനം/അർത്ഥം നാണുവസ്തുതയുടെ നിർവ്വചനം, steps, application നാണുവസ്തുതയുടെ നിർവ്വചനം. Technical terms English-നാണുവസ്തുതയുടെ നിർവ്വചനം.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► Define or explain 2 Remembering stability and fire resistance of polymers. Explain methods for improving the thermal. (3 marks)

► Define or explain 2 Remembering stability and flame resistance of polymers. Describe the properties and high temperature. (3 marks)

► Define or explain applications of polymers-PTFE, Aromatic 6 Understanding Polymers -Poly phenylene Sulphide, PPO. Explain the properties and applications of. (3 marks)

► Define or explain PEEK, Aramid fibres, Heterocyclic 5 Understanding polymers-PBI High temperature and fire-resistant polymers improving low performance polymers for high temperature use - polymers for low fire hazards - polymers for high temperature resistance. Factors affecting the thermal stability and fire resistance of polymers - LOI- Definition Properties and applications of PTFE, Aromatic Polymers like Poly phenylene Sulphide, PPO, PEEK, Aramid, Heterocyclic polymers - polybenzimidazole (PBI) Describe the structure, properties and application of polymers used in. (3 marks)

Module 2

► Define or explain 2 Remembering conducting mechanism. Describe the structure, properties and application of conducting polymers like. (3 marks)

► **Define or explain 5 Understanding Polyacetylene, poly aniline, poly paraphenylene Describe the properties and application of. (3 marks)**

► **Define or explain photo conducting and piezo electric polymers - 4 Understanding Poly vinyl carbazole , PVDF Explain the properties and application of Liquid crystalline polymers and Photo. (3 marks)**

► **Define or explain 4 Understanding resists polymers for semiconductor application. Polymers with electrical and electronic properties Conducting polymers -definition conducting mechanisms Properties and application of Polyacetylene, Polyparaphenylene, and poly aniline. Photo conducting polymers - structure, properties application of Poly vinyl carbazole Piezo electric polymers properties - structure, properties and application PVDF Photo resist polymers for semiconductor fabrication - Liquid crystalline polymers. Explain the polymers used in construction, biomedical, aerospace and. (3 marks)**

Module 3

► **Define or explain Explain the polymer concrete, types. 3 Understanding Describe the polymers and its properties. (3 marks)**

► **Define or explain 5 Understanding used for biomedical application. Explain the properties of polymers used as. (3 marks)**

► **Define or explain 3 Understanding binders for rocket propellants. Explain polymers used for Optical fibre for. (3 marks)**

► **Define or explain 3 Understanding telecommunication cables (POFC). Polymer concrete- preparation properties, application, - PCC, PIC, PMC. Polymers for biomedical applications - PHEMA, SILICONE - properties, application Polymeric binders for rocket propellants - HTPB - properties Polymers used in Optical fibre Cable(POFC) - manufacturing, properties**

and application Explain the ionomers, hydrogel, Biodegradable polymers, nano polymers. (3 marks)

Module 4

► **Define or explain 2 Remembering electrolytes and hydrogel. Describe the Synthesis, physical. (3 marks)**

► **Define or explain properties and applications of ionomers 4 Understanding based on PS, PTFE Explain the properties and applications of. (3 marks)**

► **Define or explain 4 Understanding biodegradable polymers, PLA, Chitosan. Generalize the idea of nano polymers,. (3 marks)**

► **Define or explain preparation, properties and application of 4 Understanding nano composites Properties of ionic polymers Ion - exchange, Hydrophilicity. Poly electrolytes- definition Synthesis, physical properties and applications of ionomers based on PS, PTFE, Biodegradable polymers - Properties and applications of PLA, chitosan. Polymer Nano composite - Definition, properties and application of nano composites Text / Reference: T/R Book Title/Author H.F.Mark, (Ed), Encyclopedia of polymer Science & Engineering, John Wiley & T1 Sons, New York, 1989. R1 Matrin.T.Goosey, Plastics for Electronics, Elsevier, Applied Science, 1985. Manas Chanda, Salil.K.Roy, Plastics Technology Hand book, 2nd edition, Marcel R2 Dekker, New York, 1993 R3 R.W. Dyson, Specialty Polymers, Chapman & Hall, 2nd edition, 1998. Joseph H Koo, Fundamentals properties and applications of polymer nano R4 composites, Cambridge University Press, 2016. Online Resources: SI.No Website Link 1 <https://www.researchgate.net/> 2 <https://onlinelibrary.wiley.com> 3 <https://www.ncbi.nlm.nih.gov>. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **2 Remembering stability and fire resistance of polymers. Explain methods for improving the thermal.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **2 Remembering conducting mechanism. Describe the structure, properties and application of conducting polymers like.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Explain the polymer concrete, types. 3 Understanding Describe the polymers and its properties.**

Module 4

1-mark definition: State the meaning of **2 Remembering electrolytes and hydrogel. Describe the Synthesis, physical.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://www.researchgate.net/> <https://onlinelibrary.wiley.com>
<https://www.ncbi.nlm.nih.gov>

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 5093B. Verify institutional updates against the current official curriculum.